

20-1, 10-1 (Q-1) wfmag 2k

21/06/26

* Pattern Recognition and Machine Learning

~ Christopher M. Bishop

* Pattern Classification

~ R. Duda et al

* Pattern Recognition statistical, Structural and Neural Approaches

~ R. Shalkoff

* Make a computer understand senses

* Pattern: An entity, vaguely defined, that ~~could~~ could be a name
↳ opposite of chaos

* Pattern Recognition: study of how machines can:

i) observe the environment

ii) distinguish patterns

iii) make actions

☐ Class : Similar objects

↳ Elements are representative

* Intra-class Variability → difference between same representative
 * Inter-class Similarity → similarity " diff classes

☐ 2-Class Classifier/Binary Classifier

↳ Self Explanatory
 ↳ Between two classes

☐ Representation of Pattern:

* Features: measurable quantities from patterns

Feature Vector: $\underline{x} = [x_1, \dots, x_n]^T \in \mathbb{R}^n$

* We make a decision boundary to make clusters of features

☐ Issues of Pattern Recognition:

- i) How features are generated
- ii) What are ~~bad~~ features
- iii) What type of ~~mod~~ classifier
- iv) How good is classifier

☐ Method (one with which has to describe it)

i) Preprocessing → Remove Noise from dataset
↳ unnecessary elements in data

ii) Feature Extraction → Identifiers

iii) Classification → The main process

☐ Cost of Error: Cost of misclassification → risk

C-9, 10-1 (Q-1)

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* Increasing features improves ~~statistical~~ performance

but it's not always the case as:

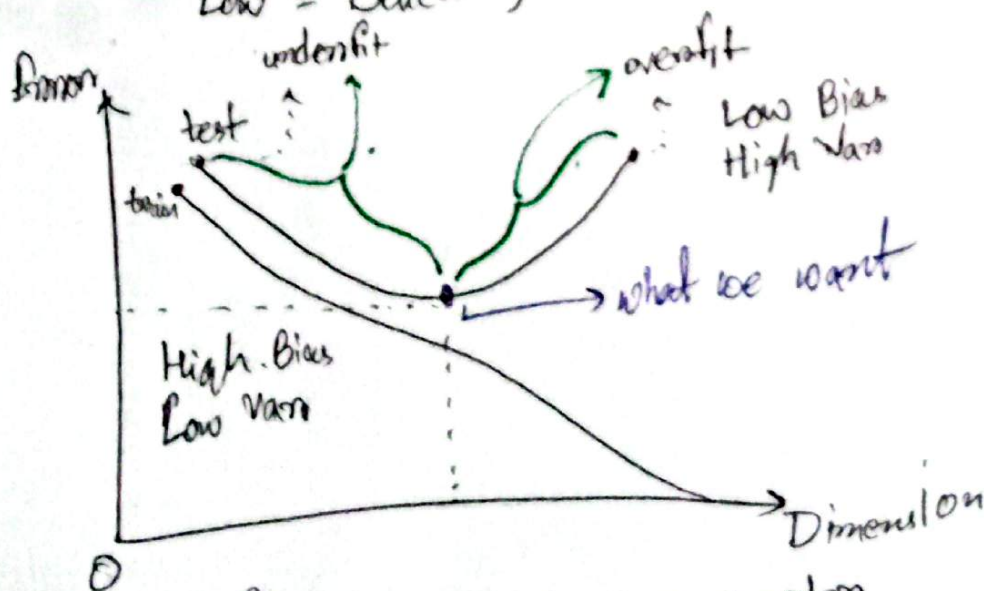
- i) unreliable features
- ii) correlations
- iii) measurement costs
- iv) noise

* High dimensions cause problem in calculations $\rightarrow$ curse

Overfitting vs Generalization:

* Bias: $\left. \begin{array}{l} \text{High} = \text{Underfit} \\ \text{Low} = \text{Overfit} \end{array} \right\}$ Better

* Variance: $\left. \begin{array}{l} \text{High} = \text{Overfit} \\ \text{Low} = \text{Better} \end{array} \right\}$



* Address Overfitting:

- i) Tuning hyperparameters
- ii) Reducing numbers of features
- iii) Using ensemble method
- iv) Using regularization

Model Evaluation

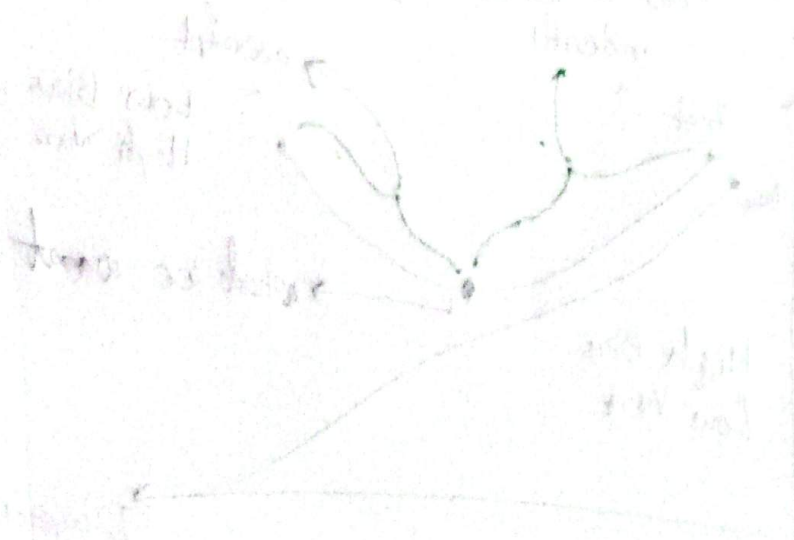
* training may yield varying results depending on splits

∴ we use folds → think outliers

↓
* ^{Multiple} Folds: Splits depending of training and testing of the same data with different parts

↓
almost perfect → but time consuming

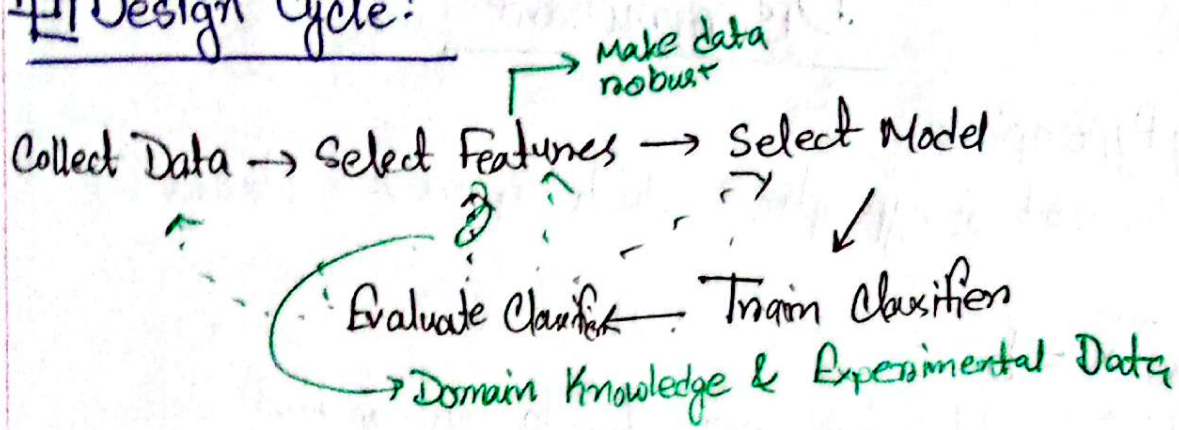
Pattern Recognition Systems



C-4, W-2(Q-1)

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Design Cycle:



* Details are in slide
They are very self explanatory
to the title.

If for some reason you don't understand
what the words mean,
Pls go learn English again, future me.

Discriminant

☐ Hypersplane:

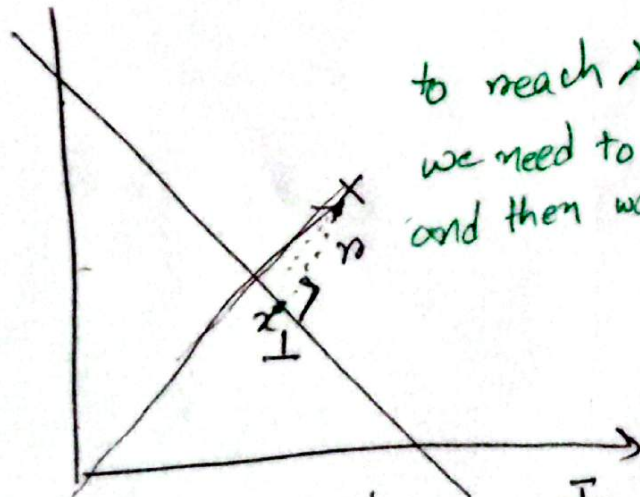
* We want a hypersplane which separates classes

* $w^T x = b$

↓ $\hookrightarrow$ normal vector $\rightarrow \perp$ to hypersplane

$y(x) = w^T x + w_0 \rightarrow \begin{bmatrix} w_0 \\ w_1 \\ \vdots \end{bmatrix}^T \cdot \begin{bmatrix} 1 \\ x_1 \\ \vdots \end{bmatrix}$

* Unit vectors are measuring unit



to reach x ,
we need to cross $x_{\perp}$
and then walk n units

$$x = x_{\perp} + n \cdot \frac{w}{\|w\|}$$

$$\text{or, } w^T x = w^T x_{\perp} + n \cdot \frac{w^T w}{\|w\|}$$

$$\text{or, } w^T x = w^T x_{\perp} + n \|w\|$$

$$\text{or, } w^T x + w_0 = w^T x_{\perp} + n \|w\| + w_0$$

$$\text{or, } y(x) = 0 + n \|w\|$$

$$\therefore n = \frac{y(x)}{\|w\|}$$

C-5, w-2(a-1)

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+ another way of calculation:

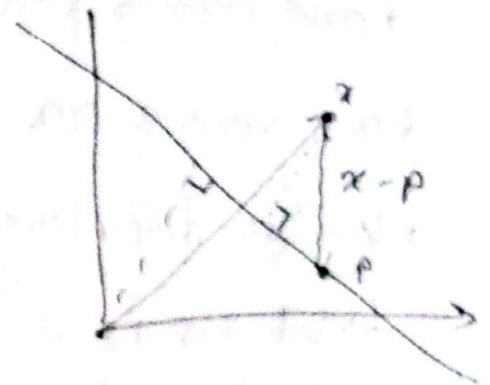
Let, we take p on any decision boundary.

$$\therefore y(p) = 0$$

$$\therefore w^T p + w_0 = 0 \quad (i)$$

$$\therefore w^T p = -w_0 \quad (ii)$$

$$\therefore \text{Bridge Vector} = (x - p)$$



$$\therefore n = (x - p)^T \left(\frac{w}{\|w\|} \right)$$

$$= \frac{w^T x - w^T p}{\|w\|}$$

$$= \frac{w^T x + w_0}{\|w\|}$$

$$\|w\| = \sqrt{w_1^2 + w_2^2 + \dots}$$

$$\therefore y(x) = n = \frac{y(x)}{\|w\|} \rightarrow \text{signed distance} \rightarrow \begin{matrix} \text{+ve} \\ \text{-ve} \end{matrix} \text{ matters}$$

$$\therefore \text{for origin: } n = \frac{w^T x + w_0}{\|w\|} \quad \left\{ \begin{array}{l} x=0 \text{ at origin} \\ w_0 \end{array} \right\} = - \frac{w_0}{\|w_0\|}$$

C-6, W-2, (Q-1)

01/07/26

Multiple Classification:

* one versus rest

* one versus one $\rightarrow$ separate C_i and C_j for n classes

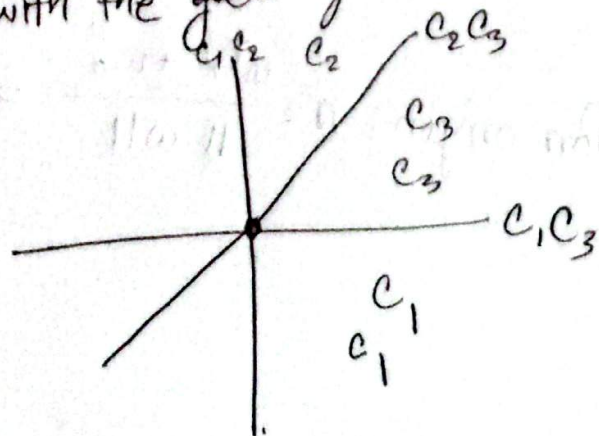
* k -class discriminant: $y_{C_i}(x) = w_{C_i}^T x + w_{C_i,0}$

$\hookrightarrow$ but this causes an issue because the straight lines might create one-versus-one again

$\therefore$ we use $w_j^T x + w_{j,0} = w_k^T x + w_{k,0}$

to derive $(w_j - w_k)^T x + (w_{j,0} - w_{k,0}) = 0$

$\rightarrow$ we take lines and use k on 1v1 discriminant to get regions $\rightarrow$ gives three classes and the one with the get highest vote is chosen



Convex vs Non Convex:

* If ~~two~~ a shape can cut shapes then it is convex



If $y_1(x) > y_2(x)$ holds true for all cases,
then its convex:

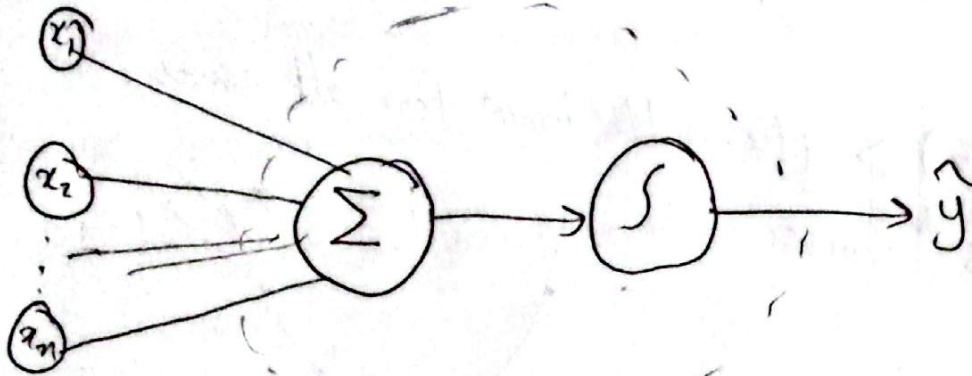


C-7, W-3 (Q-2)

05/07/20

Perceptron: just a neuron

↳ Frank Rosenblatt (1958)



$$y(x) = f(\omega^T \phi(x)) \rightarrow \hat{y} = s(\omega^T \cdot x)$$

$$\omega^T \phi(x) t_n > 0$$

[check derivations from slide]

$$\downarrow$$
$$t_n = -1, 1$$

Perceptron Criterion (Error Function)

$$E_p(\omega) = - \sum_{n \in \text{EM}} \omega^T \phi_n t_n$$

Taking the derivative:

$$\nabla E_p^n(\omega) = - \phi_n t_n \rightarrow \text{learning rate, } \alpha$$

$$\omega^{(n+1)} = \omega^n - \eta \nabla E_p^n(\omega)$$

C-8, W-3, (Q-1)

06/07/26

[a bit late]

→ stochastic, $\nabla E_p(w) = -\phi_n t_n$

→ for bigger dataset & streaming dat

Batch Update: → for smaller dataset & convergence

$$\nabla E_p(w) = - \sum_{n \in M} \phi_n t_n$$

$$w^{T+1} = w^T + \eta \sum_{n \in M} \phi_n t_n$$

→ strictly positive

* New error = Old Error - $\frac{1}{5th}$

→ this implies that the error will always be smaller than before

$\phi(x)$ → transforming feature vector

→ used to make things linearly separable

* Hand crafted → specific domain data

Least Squared Method: make a line to make distances smaller

$$SSE = \sum_{i=1}^n (y - \hat{y})^2$$

Let,

$$y(x) = w_0^T x + w_0 \rightarrow [w_0, w_1, w_2, \dots] \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \end{bmatrix}$$

$$= \tilde{w}^T \tilde{x}$$

C-9/w-3, Q-1

08/07/26

* $SSE = \sum_{i=1}^n (y - \hat{y})^2$

→ for multiple → $E_D(\tilde{W}) = \frac{1}{2} \text{Tr} \left\{ (\tilde{X} \tilde{W} - T)^T (\tilde{X} \tilde{W} - T) \right\}$

Trace function
take diagonal

* Total Error = $\sum_{n=1}^N \sum_{k=1}^K (e_{nk})^2$

error of classifying nth element in kth class

* To reduce error, we take first derivative

[do math from slide → Chap 2. pptx Page 28]

I don't get paid enough for this

* $(\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \rightarrow \tilde{X}^+ \rightarrow$ pseudo inverse

$\therefore W = \tilde{X}^+ T$

$\therefore y(x) = \tilde{W}^T \tilde{x} = T^T (\tilde{X}^+)^T \tilde{x}$

Side Math (Page - 30)

(i) $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix}$

for w_0 hours of study

(ii) $T = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

Fail Pass

(iii) $(X^T X)^{-1} = \frac{1}{14} \begin{bmatrix} 21 & -7 \\ -7 & 3 \end{bmatrix}$

(iv) $(X^T X)^{-1} T = \frac{1}{14} \begin{bmatrix} 21 \\ 7 \end{bmatrix}$

$(X^T X)^{-1} X^T T = \frac{1}{14} \begin{bmatrix} 14 & 7 & -7 \\ -4 & -1 & 5 \end{bmatrix} = \tilde{W}$

(v) Say $X^n = \begin{bmatrix} 1 & 3 \end{bmatrix}$

$T^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(vi) $(X^+)^T = \frac{1}{14} \begin{bmatrix} 14 & -4 \\ 7 & -1 \\ -7 & 5 \end{bmatrix}$

(vii) $y(x) = T^T (X^+)^T \tilde{X}$

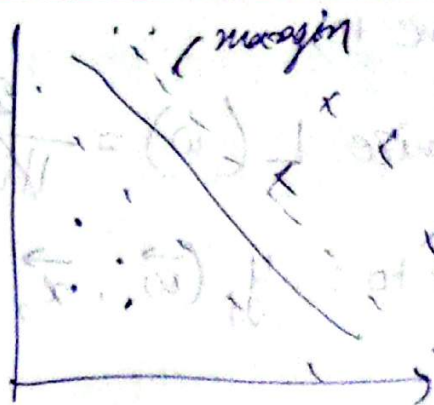
C-2, w-5(a-1.5)

13/07/26

$$+ \text{margin} = \frac{2}{\|\vec{w}\|}$$

$$y_i = 1$$

$$y_i = \begin{cases} 1 & \text{if } \vec{w} \cdot \vec{x}_i + b \geq 1 \\ -1 & \text{if } \vec{w} \cdot \vec{x}_i + b \leq -1 \end{cases}$$



can be written as, $y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1$

we bring in Lagrange for this optimization

2.1 time bylec

$$L(x, \lambda) = f(x) - \lambda g(x)$$

where $g(x) \geq 0$

Primal: $\min_x \max_{\lambda \geq 0} L(x, \lambda)$

but working with this hard
we need to make it

we are trying to minimize
and violating it creates a
bigger number

Dual: $\max_{\lambda \geq 0} \min_x L(x, \lambda)$

equivalent to
work with x first

We can't usually swap
the min max orders though

However, KKT proposed 4 conditions that state
that if true then primal and dual will provide same
result

- i) $\nabla f(x) - \lambda \nabla g(x) = 0$
- ii) $g(x) \geq 0$
- iii) $\lambda \geq 0$
- iv) $\lambda g(x) = 0$

We have to,

$$\text{minimize: } L(\omega) = \frac{2}{\|\vec{\omega}\|} \frac{\|\vec{\omega}\|^2}{2}$$

$$\text{subject to: } y_i (\vec{\omega} \cdot \vec{x}_i + b) \geq 1$$

∴ with Lagrange,

$$L_p = \frac{\|\vec{\omega}\|^2}{2} - \sum_{i=1}^N \lambda_i (y_i (\vec{\omega} \cdot \vec{x}_i + b) - 1)$$

$\downarrow$ $\downarrow$ $\downarrow$
 $f(x)$ λ $g(x)$

(*) $g(x) - (x) \lambda = 0$ for all

New Constraints:

$$\frac{\partial L_p}{\partial \omega} = 0 \longrightarrow \vec{\omega} = \sum_{i=1}^N \lambda_i y_i \vec{x}_i$$

$$\frac{\partial L_p}{\partial b} = 0 \longrightarrow \sum_{i=1}^N \lambda_i y_i = 0$$

By replacing $\vec{\omega}$ in L_p ,

$$L_p = \frac{\|\vec{\omega}\|^2}{2} - \sum_{i=1}^N \lambda_i y_i \vec{\omega} \cdot \vec{x}_i - \sum_{i=1}^N \lambda_i y_i b + \sum_{i=1}^N \lambda_i$$

we have to minimize it

$$L_D = \sum_{i=1}^N \lambda_i \frac{\vec{\omega} \cdot \vec{\omega}}{2} = \left[\sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j y_i y_j \vec{x}_i \cdot \vec{x}_j \right]$$

- Steps $\rightarrow$
- i) find $z_i \leftarrow$ will be provided in exam
 - ii) Use z_i to find w and $b \rightarrow$ avg of b_i
 - iii) Use eqⁿ to find answers

* Points on the margin are called support vectors

(3) additional loss

$$3 - 1 \leq (d + w \cdot w)$$

minimize: soft-margin

$$\min_{w, b} \sum_{i=1}^n \xi_i + \frac{1}{2} \|w\|^2$$

margin loss soft margin

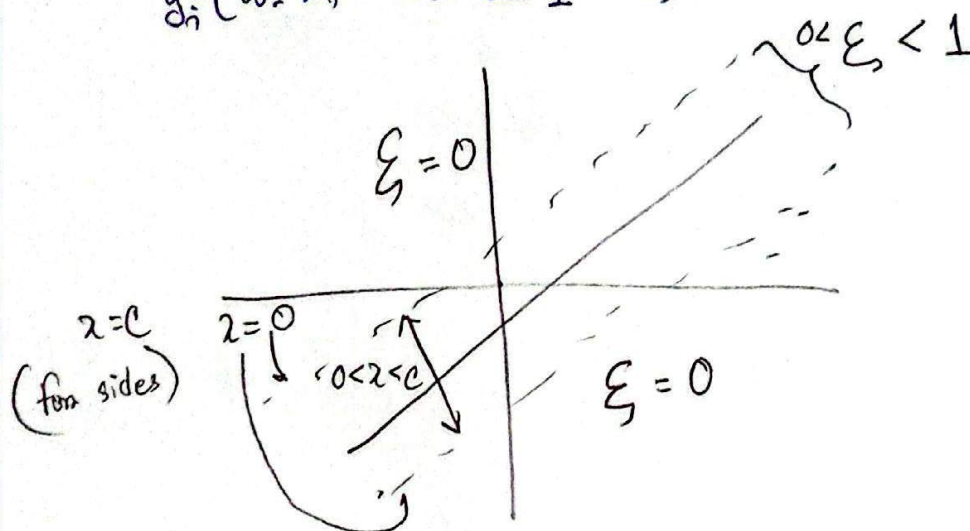
* But what if outliers? } SVM fails here
 & What if outliers on other class? }
 To remedy this, we introduce soft margin.

Soft Margin SVM

* Slack Variable (ξ):

Relaxes the margin with some penalty

$$y_i (w \cdot x_i + b) \geq 1 - \xi$$



∴ Primal Objective:

$$\min_{w, b, \xi} \underbrace{\frac{1}{2} \|w\|^2}_{\text{Max this}} + C \underbrace{\sum_{i=1}^N \xi_i}_{\text{Min this}}$$

Non-Linear SVM:

$$L_D = \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum \lambda_i \lambda_j y_i y_j x_i x_j$$

$$\downarrow$$
$$= \sum_{i=1}^N \lambda_i - \frac{1}{2} \sum_{i,j} \lambda_i \lambda_j \phi(x_i) \phi(x_j)$$

but this increases computation power
to ~~reduce~~ reduce this computation power, we use kernel trick
→ But, we need a kernel trick that utilizes the old dot product

Bayesian Decision Theory

* Bayes Theory: $P(H|E) = \frac{P(E|H)P(H)}{P(E)}$

$\uparrow$ likelihood $\rightarrow$ prior
 $\uparrow$ posterior $\rightarrow$ evidence $\rightarrow$ Believe

$$P(w_i|x) = \frac{P(x|w_i) \cdot P(w_i)}{P(x)}$$

* likelihood $\rightarrow$ chances of ^{an evidence} something happening
 given ~~an~~ evidence something happening

□ Error: $P(\text{error}|x) = 1 - \text{Probability of being correct}$

$$P(\text{error}|x) = \begin{cases} P(w_1|x) & \text{if we decide } w_2 \\ P(w_2|x) & \text{" " " " } w_1 \end{cases}$$

$\rightarrow$ this is because the error depends on what is the wrong probability

$$\therefore P(\text{error}|x) = \min [P(w_1|x), P(w_2|x)]$$

$$P(\text{error}) = \int_{-\infty}^{\infty} P(\text{error}, x) dx = \int_{-\infty}^{\infty} P(\text{error} | x) \cdot p(x) dx$$

↳ we have to minimize this

* For Bayesian,

w_1 , if $P(x | w_1) P(w_1) > P(x | w_2) P(w_2)$ } is computationally efficient
else w_2

w_1 , if $P(w_1 | x) > P(w_2 | x) \rightarrow$ normal use
else w_2

w_1 , if $P(x | w_1) / P(x | w_2) > P(w_2) / P(w_1) \rightarrow$ for sensors x
else w_2

Relative Frequency (Objective) [Probability]

→ using experiment to find possible values

~~not find prob~~

1. Get data

2. Find $P(C_i) \rightarrow \frac{\text{no. of class val}}{\text{Total No. of obj}}$

3. Discretize → i) Bin them

ii) Probability of each bin

↳ that $P(x = \text{bin} | C_i)$

↳ $\frac{\text{No. of class val in bin}}{\text{Total value of class}}$

4. $P(x) = P(x | C_1) + P(x | C_2) + \dots$

5. $P(C_i | x) = \frac{P(x | C_i) P(C_i)}{P(x)} \rightarrow$ This finds probability
↳ without $P(x)$ is too making decision

Subjective Approach: when no experiment
↳ we assume things

1. We assume $P(x|c_i)$ and $P(c_i)$ are from distributions

C-16, W-6, Q-2.5

29/07/26

* We take α action with w categories } $Loss = \lambda(\alpha | w)$
↳ loss if α is taken for w classification

* We have to make a decision rule that selects α_i so that λ_i is least

$$\text{Risk, } R(\alpha_i | x) = \sum_{j=1}^c \lambda(\alpha_i | w_j) P(w_j | x)$$

↳ Risk of taking α_i action

Slide Calc

$$\begin{aligned} \therefore R(\alpha_1 | x) &= \lambda_{11} P(w_1 | x) + \lambda_{12} P(w_2 | x) \\ &= 85 \cdot 0.85 \end{aligned}$$

$$\therefore R(\alpha_2 | x) = 15$$

Zero-One Loss Function:

↓
loss is equal for all decision $\rightarrow 0$ for correct
1 " incorrect

$$R(\alpha_i | x) = \sum_{j=1}^c 2(\alpha_i | \omega_j) P(\omega_j | x)$$

$$= \sum_{i \neq j} P(\omega_j | x) = 1 - P(\omega_i | x)$$

Multi-Class Category: $g_i(x) > \text{all} \rightarrow i \text{ category}$

Loss for i decision taking
↓ considering x
Loss: $g_i(x) = -R(\alpha_i | x)$
↳ mathematical convenience for maximizing

for zero-one: $g_i(x) = P(\omega_i | x) - 1$
Dropping constant

$$g_i(x) = P(\omega_i | x)$$

$$= \frac{P(x | \omega_i) P(\omega_i)}{P(x)}$$

$$= P(x | \omega_i) P(\omega_i)$$

$\therefore g_i(x) = \ln P(x | \omega_i) + P(\omega_i)$ ← function just scales
taking $\ln$ to as it
is a monotonically
increasing function

MLE

$$P(w|x) = \frac{P(x|w)P(w)}{P(x)}$$

$$P(w|x) \propto P(x|w)P(w)$$

└─────────> Priors

* We want to find best μ and σ

* We move μ across point to find the highest σ MLE

We use σ to get the curve

* we have to estimate the number of parameters

* We assume it fits Gaussian Distribution

└─> why? don't ask

i) Maximum Likelihood Estimation ──> fixed but unknown

└─> ii) Bayesian Estimation ──> prior distributive

[Check Slide]

Idk how to write any of this

~~on what any of this even means~~

Maximum Likelihood Estimation:

Likelihood of $\theta \rightarrow p(D|\theta) = \prod_{k=1}^n p(x_k|\theta) \leftarrow \text{Not Conditional of } \theta$

$\uparrow$
 Probability of D
 parameterized by θ

$\downarrow$
 also $p(D, \theta)$

Goal $\rightarrow$ find $\hat{\theta}$ to max $p(D|\theta)$

+ we take log for a smooth curve

$$\hat{\theta} = \arg \max_{\theta}$$

$$\hat{\mu} = \arg \max_{\mu} \sum_{k=1}^n \left(-\ln(\sqrt{2\pi}\sigma) - \frac{(x_k - \mu)^2}{2\sigma^2} \right)$$

$$\frac{d}{d\mu}$$

$$\hat{\mu} = \frac{1}{n} \sum_{k=1}^n x_k = \text{avg}(x)$$

$$\sigma^2 = \frac{1}{n} \sum_{k=1}^n (x_k - \mu)^2$$

I am not doing derivation

Random Variable: ~~Numerical~~ Variable belonging to the numeric representation of a sample space

$$X: S \longrightarrow \mathbb{R}$$

$\downarrow$
sample space

Expected Value: Long run average of a random variable

Discrete: $E(X) = \sum x_i \cdot P(X=x_i)$

Continuous: $E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$

Biased and Unbiased Estimator:

* Biased $\longrightarrow$ if expected value $\neq$ true value $E(\hat{\theta}) \neq \theta$

* Unbiased $\longrightarrow$ if expected value $=$ true value $E(\hat{\theta}) = \theta$

$\therefore \sigma_\mu$ is unbiased

σ^2 is biased $\rightarrow \therefore$ we need to adjust it a bit

$\hookrightarrow$ for larger datasets,
this bias is negligible

M A P

Maximum A Priori Posteriori

- * Using Prior knowledge for better estimation
- * We assume parameters are also distributed
- * just check the slide box,
there is honestly nothing to write

$$\theta = \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2} \underset{\substack{\uparrow \\ \text{data}}}{x} + \frac{\sigma^2}{\sigma_0^2 + \sigma^2} \underset{\substack{\uparrow \\ \text{parameter}}}{\theta}$$

* MAP = MLE when $\sigma_0 \rightarrow \infty$
 $\hookrightarrow$ variance of parameter

* MAP works poorly when prior $(p(\theta))$ is bad

C-21, W-9, Q-3

23/06/26

MLE & MAP:

Given

$$P(\text{small ears} | \text{Cat}) = 0.8$$

$$P(\text{small ears} | \text{Dog}) = 0.1$$

$$\therefore P(\text{Large ears} | \text{Cat}) = 0.2$$

$$\therefore P(\text{Large dog} | \text{Dog}) = 0.9$$

$$i) \therefore P(\text{Dog} | \text{Large Ears}) = \frac{P(\text{Large Ears} | \text{Dog}) P(\text{Dog})}{P(\text{Large Ears})}$$

$$= \frac{P(\text{Large Ears} | \text{Dog}) P(\text{Dog})}{P(\text{Large Ears} | \text{Dog}) P(\text{Dog}) + P(\text{Large Ears} | \text{Cat}) P(\text{Cat})}$$

ii)

#2

$$p(z | c_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(z - \mu_k)^2}{2\sigma_k^2}} \quad \leftarrow \text{MLE}$$

$$p(z | c_k) p(c_k) \leftarrow \text{MAP}$$

for decision boundary $\rightarrow p(z | \text{salmon}) = p(z | \text{bass})$

PCA

Curse of Dimensionality: Feature is not always good

* Feature increase = Data increase
Otherwise, it can create sparse graphs
↳ which creates errors

To remove this we use PCA:

PCA: reducing d dimension to p

Worst: $d = p$ $p = d$

Best: $p < d$

↓
we find what is
plot the data in axes and
find which data is more
important.

Next Sunday quiz (maybe)

Main idea:

- i) Variation gives information
- ii) Combine
- iii) ^{Zero} Redundancy

- Steps:
- i) Shift graph that gets centers at
 - ii) Get a line that fits data $\rightarrow$ The line that maximizes $\sum d_i^2$
 $\downarrow$
 PC1
 - iii) Slope indicates how much of each feature is important
 - iv) PC2 is orthogonal to PC1

$$\frac{SS(\text{distances for PC-1})}{n-1} = \text{Variation for PC1}$$

* We use scree plot to determine what to drop